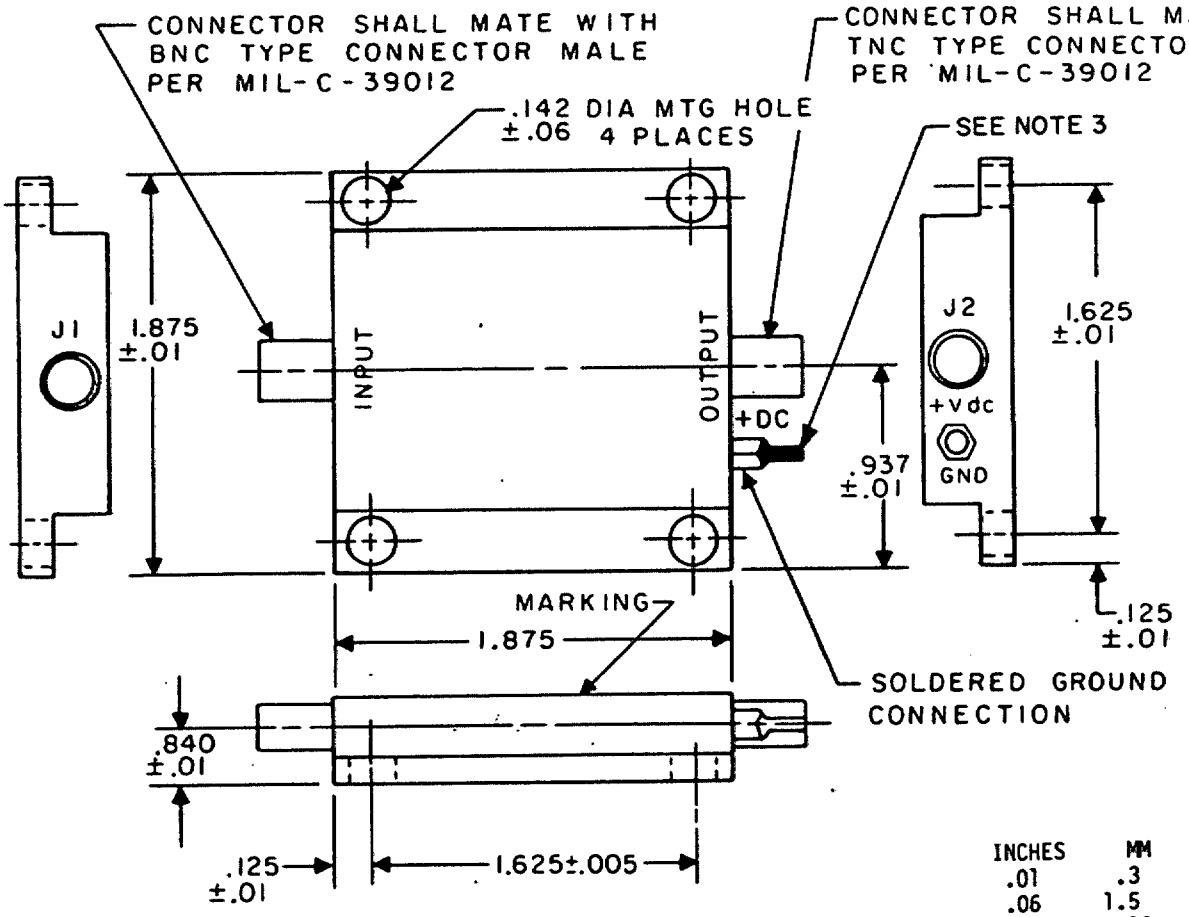
MILITARY SPECIFICATION SHEET

AMPLIFIERS, RADIOFREQUENCY AND MICROWAVE,
SOLID-STATE, LOW NOISE, WIDEBAND, COAXIAL CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring amplifiers described herein shall consist of this specification sheet and the latest issue of MIL-A-28875.



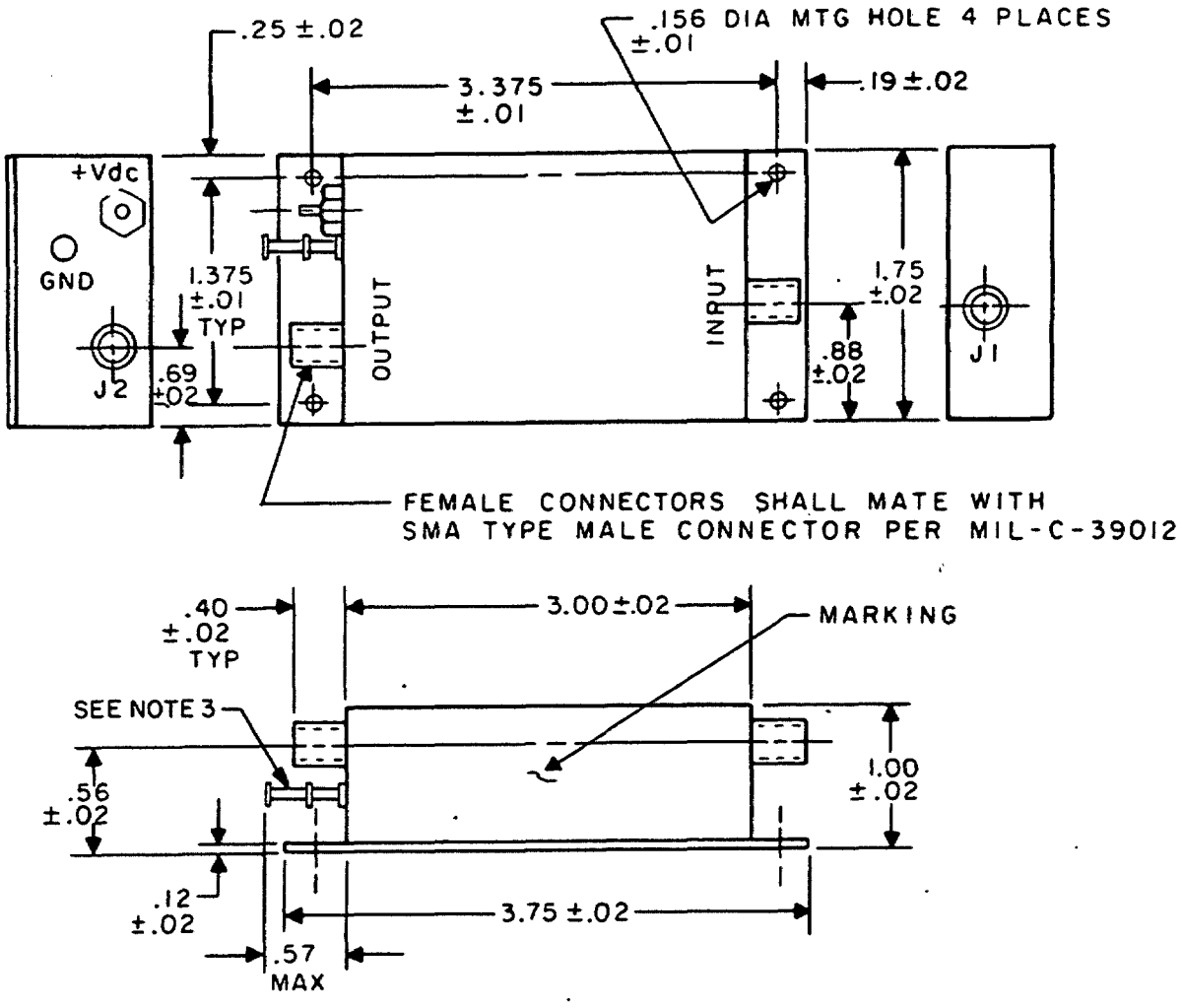
INCHES	MM
.01	.3
.06	1.5
.125	3.18
.142	3.60
.840	21.33
.937	23.80
1.625	41.27
1.875	47.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 1. Dimensions and configuration, dash number -01.

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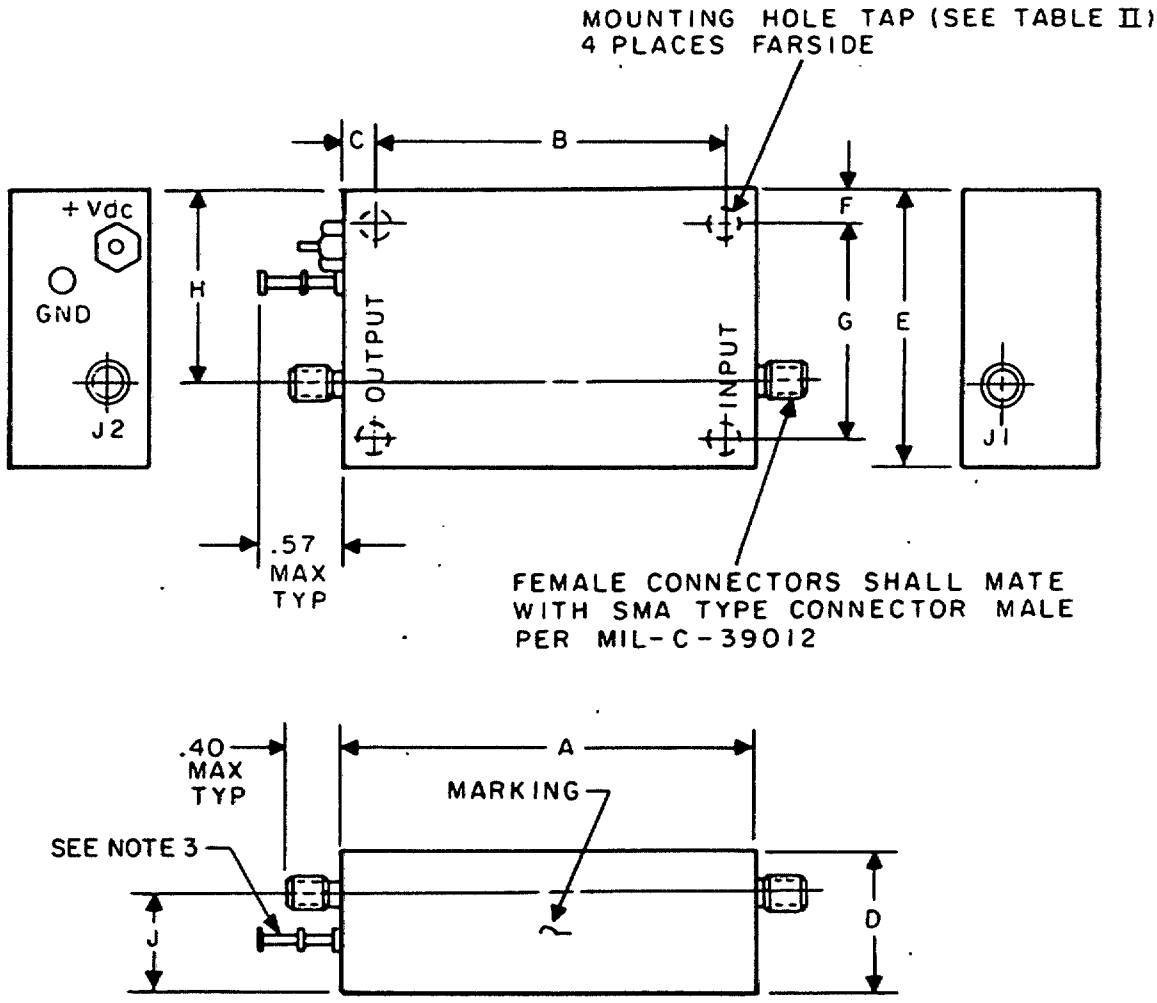


INCHES	MM	INCHES	MM
.01	.3	.69	17.5
.02	.5	.88	22.4
.12	3.0	1.00	25.4
.156	3.96	1.375	34.93
.19	4.8	1.75	44.4
.25	6.4	3.00	76.2
.40	10.2	3.375	85.73
.56	14.2	3.75	95.2
.57	14.5		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 2. Dimensions and configuration, dash number -02.

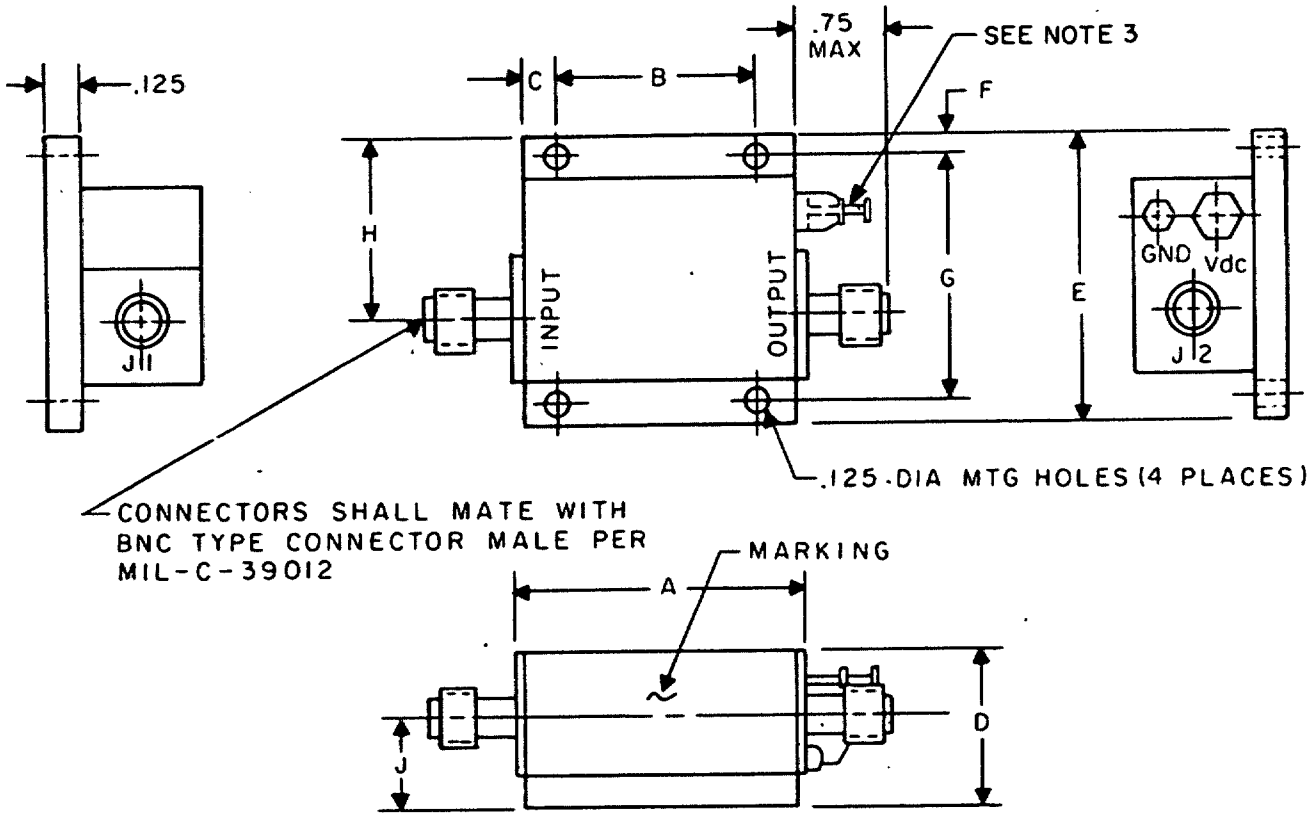
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 3. Dimensions and configuration, dash numbers -03, -04, -07, -08, -10, -11, and -12.

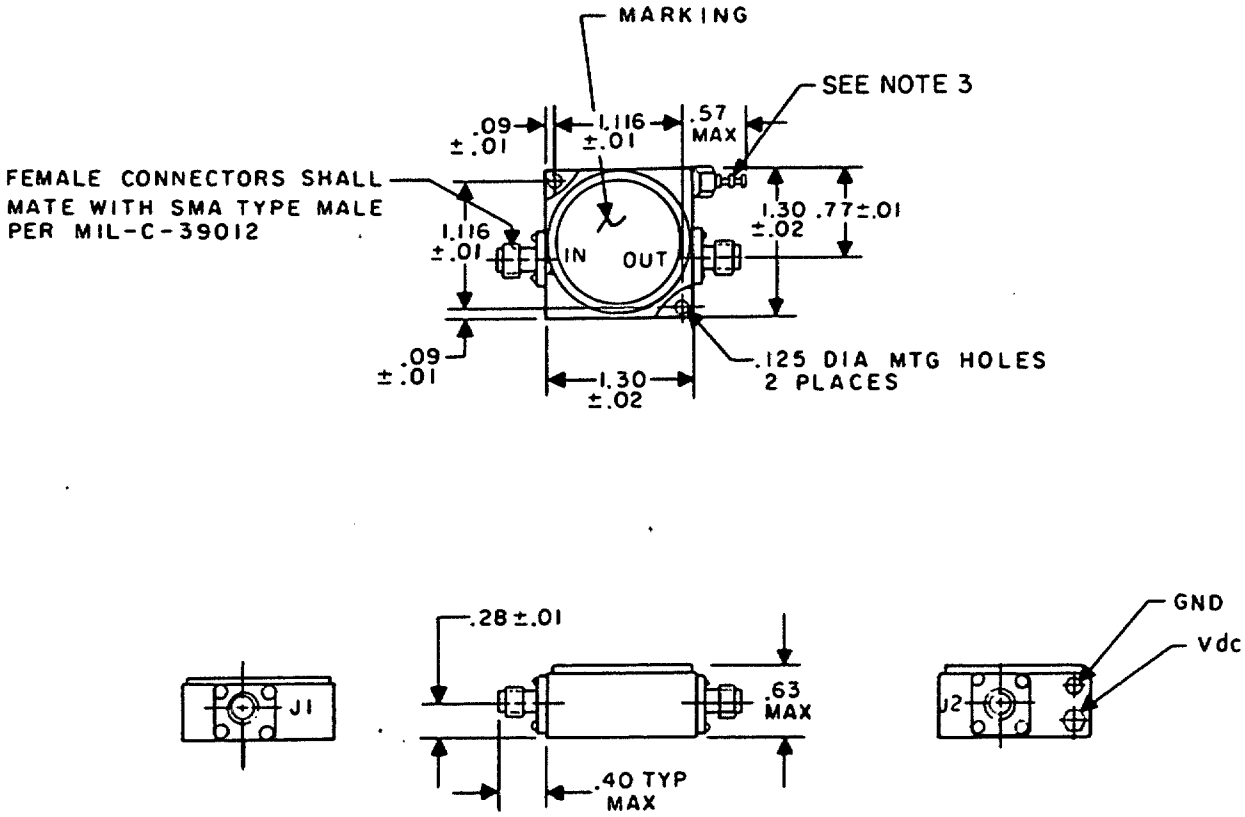
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 4. Dimensions and configuration, dash numbers -05 and -09.

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INCHES	MM
.01	.3
.02	.5
.09	2.3
.125	3.18
.28	7.1
.40	10.2
.57	14.5
.63	16.0
.77	19.6
1.116	28.35
1.30	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Barrel diameter shall accept two turns of AWG NO. 24 wire.

FIGURE 5. Dimensions and configuration, dash number -06.

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TABLE I. Electrical characteristics (25°C unless otherwise specified).

Dash number	Frequency range	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBm)	VSWR Max		Noise figure (dB)	Intercept point (dBm)	Gain vs temperature (dB)	RF input power	Isolation (dB)	Phase linearity (deg.)
		Voltage	Current Max (mA)	Min	Max	Min	IN	OUT	Max	Min		Max	Min	Max
01 S1 S2 N	5-250 MHz	15 ±1%	15	25	±0.5	+5	2.0:1	2.0:1	1.5	+17	±0.5	---	---	---
02 S1 S2 N	100-200 MHz	15 ±1%	30	27	±1.0	+5	2.0:1	2.0:1	2.5	+15	---	---	---	---
03 S1 S2 N	10-400 MHz	15 ±1%	40	40	±1.5	+6.5	2.5:1	2.5:1	3.5	+19	---	---	---	---
04 S1 S2 N	10-500 MHz	15 ±1%	130	38	±1.5	+19	2.5:1	2.5:1	4	+34	---	---	---	---
05 S1 S2 N	5-1000 MHz	15 ±1%	15	14	±1.0	-2	2.0:1	2.0:1	5	---	---	---	---	---
06 S1 S2 N	100-2000 MHz	15 ±1%	23	9	±1.0	+5	2.0:1	2.0:1	6.5	+15	---	---	---	---
07 S1 S2 N	500-2000 MHz	12 ±1%	200	36	±1.0	+15	2.0:1	2.0:1	3.5	+25	---	---	---	---
08 S1 S2 N	500-2000 MHz	12 ±1%	160	27	±1.0	+15	2.0:1	2.0:1	3.5	+25	---	---	---	---
09 S1 S2 N	500-2000 MHz	15 ±1%	110	25	±1.0	+12	2.0:1	2.0:1	4.0	+23	---	---	---	---

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TABLE I. Electrical characteristics (25°C unless otherwise specified) - Continued.

Dash number	Frequency range	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBH)	VSWR Max		Noise figure (dB)	Intercept point (dBm)	Gain vs temperature (dB)	RF input power	Isolation (dB)	Phase linearity (deg.)
		Voltage	Current Max (mA)	Min	Max	Min	IN	OUT	Max	Min		Max	Min	Max
S1 10 S2 N	8-12 GHz	12 ±1%	250	37	±2.0	+13	2.0:1	2.0:1	5.5	+23	---	---	---	---
S1 11 S2 N	8-12 GHz	12 ±1%	150	22	±1.5	+13	2.0:1	2.0:1	5.5	+23	---	---	---	---
S1 12 S2 N	12-18 GHz	12 ±1%	375	33	±2.0	+10	2.0:1	2.0:1	7.0	+20	---	---	---	---

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TABLE II. Dimensions, physical and environmental characteristics.

Dash number	Dimensions										Weight (LB) Max	Temperature range		Seal	Salt	EMI	RF connectors		Figure number
	A	B	C	D	E	F	G	H	J	K		Operating case temp	Nonoperating				Input	Output	
01 S1 S2 N	---	---	---	---	---	---	---	---	---	---	.44	-54/+71°C	-54/+85°C	Non-hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	1
02 S1 S2 N	---	---	---	---	---	---	---	---	---	---	.44	-54/+71°C	-54/+85°C	Non-hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	2
03 S1 S2 N	1.33 ±.02	.750 ±.01	.29 ±.01	.700 ±.02	1.03 ±.02	.11 ±.01	.810 ±.01	.555 ±.02	.352 ±.02	2-56 UNC-2B .15 Dp	.28	-54/+85°C	-54/+100°C	Non-hermetic	Applicable	CEO3 of MIL-STD-461	Female	Female	3
04 S1 S2 N	2.35 ±.02	1.774 ±.01	.29 ±.01	.700 ±.02	1.03 ±.02	.11 ±.01	.810 ±.01	.555 ±.02	.352 ±.02	2-56 UNC-2B x .15 Dp	.28	-54/+85°C	-54/+100°C	Non-hermetic	Applicable	CEO3 of MIL-STD-461	Female	Female	3
05 S1 S2 N	1.37 ±.02	1.000 ±.01	.19 ±.01	.84 ±.02	1.50 ±.02	.11 ±.01	1.281 ±.01	.90 ±.02	.49 ±.02	---	.24	-45/+81°C	-57/+81°C	Non-hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	4
06 S1 S2 N	---	---	---	---	---	---	---	---	---	---	.22	-54/+85°C	-54/+100°C	Hermetic	Applicable	CEO3 of MIL-STD-461	Female	Female	5
07 S1 S2 N	2.08 ±.02	1.58 ±.01	.25 ±.01	.67 ±.02	2.15 ±.02	.20 ±.01	.75 ±.01	.78 ±.02	.31 ±.02	4-40 UNC-2B x.25Dp	.17	-54/+85°C	-54/+100°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	3
08 S1 S2 N	2.08 ±.02	1.58 ±.01	.25 ±.01	.67 ±.02	1.15 ±.02	.20 ±.01	.75 ±.01	.78 ±.02	.31 ±.02	4-40 UNC-2B x.25Dp	.17	-54/+85°C	-54/+100°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	3

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TABLE II. Dimension, physical and environmental characteristics - Continued.

Dash number	Dimensions										Weight (LB) Max	Temperature range		Seal	Salt	EMI	RF connectors		Figure number
	A	B	C	D	E	F	G	H	J	K		Operating case temp	Nonoperating				Input	Output	
09	S1 S2 N	2.85 ±.02	2.450 ±.01	.20 ±.01	.85 ±.02	1.95 ±.02	.11 ±.01	1.732 ±.01	1.12 ±.02	.49 ±.02	---	.77	-54/+85°C	-54/+100°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	4
10	S1 S2 N	2.25 ±.02	1.750 ±.01	.25 ±.01	.67 ±.02	1.15 ±.02	.20 ±.01	.75 ±.01	.64 ±.02	.31 ±.02	4-40 UNC-2B x.25Dp	.22	-54/+71°C	-54/+85°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	3
11	S1 S2 N	1.75 ±.02	1.250 ±.01	.25 ±.01	.67 ±.02	1.15 ±.02	.20 ±.01	.75 ±.01	.64 ±.02	.31 ±.02	4-40 UNC-2B x.25Dp	.15	-54/+71°C	-54/+85°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	3
12	S1 S2 N	4.42 ±.02	4.25 ±.01	.09 ±.01	.54 ±.02	1.26 ±.02	.09 ±.01	1.09 ±.01	.65 ±.02	.23 ±.02	2-56 UNC-2B x.25Dp	.44	-54/+71°C	-54/+85°C	Hermetic	N/A	CEO3 of MIL-STD-461	Female	Female	3

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REQUIREMENTS:

Dimensions and configuration: See figures and table II

Electrical characteristics: See table I.

Physical and environmental characteristics: See table II.

Part number: M28875/6-(dash number from tables I and II).

NOTE: Amplifiers covered by this specification sheet are substitutable for the following manufacturer's or user's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	Manufacturer's code	Manufacturer's part number
01S	82918	380-10266-1
01S	55791	W250D-1
02S	24539	AK-200
03S	14482	6200-375
04S	14482	6200-384
05S	80063	SM-C-800833
05S	24539	UTA-8201M
06S	24539	AMT-2002
06S	80063	SM-A-874285-2
07S	24539	AWT-2034
08S	24539	AWT-2033
09S	24539	AWT-2045
10S	24539	AMT-12435
11S	24539	AMT-12433
12S	14482	5320-316

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0968-6)

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